

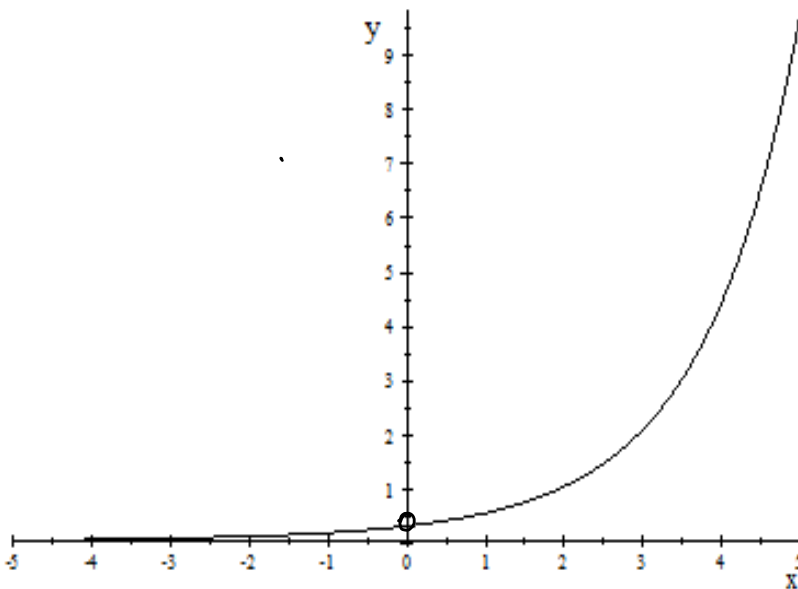
Unit 2.2 The limit of a function

pp 83-93.

Note Title

2015/02/23

Look at $f(x) = \frac{e^x - 1}{3x}$

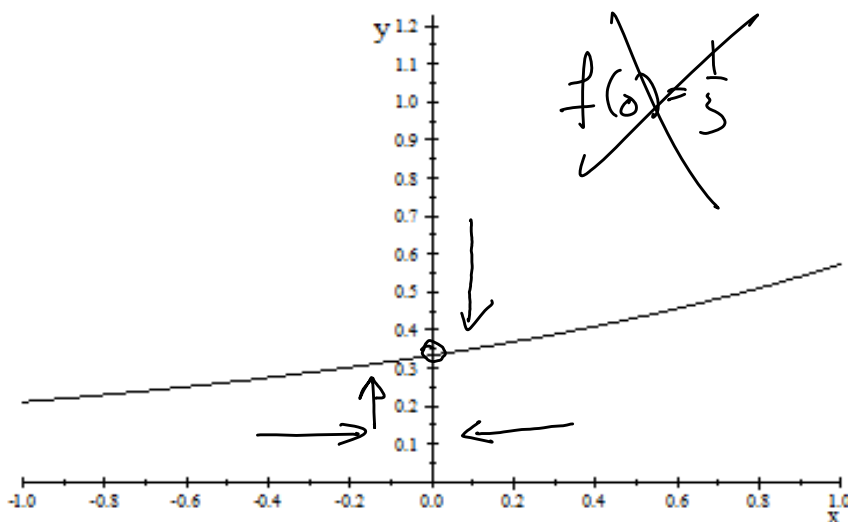


The function is defined everywhere except in $x=0$.

Yet the graph gets close to a specific value as x is close to 0.

$$f(x) = \frac{e^x - 1}{3x}$$

Have a closer look:



$f(0)$ is undefined

$$f(0,01) = 0,335$$

$$f(0,0001) = 0,33338$$

We say that

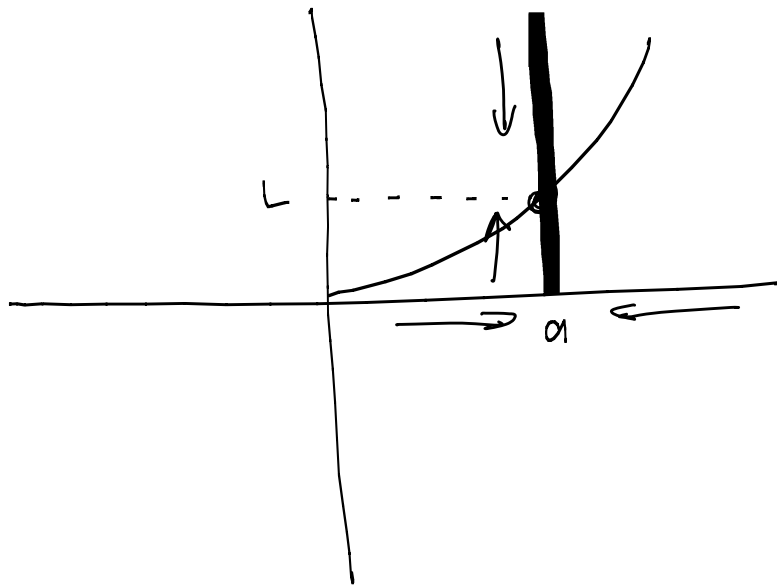
$$\lim_{x \rightarrow 0} \frac{e^x - 1}{3x} = \frac{1}{3}$$

As x gets closer to 0, y gets closer to 0,3333...

Definition : Suppose f is defined near the number a .
 (f is defined in an open interval around a , not necessarily in a itself)

Then we write $\lim_{x \rightarrow a} f(x) = L$

if we can make the value of f as close to L as we like by taking x close enough to a .



$$\lim_{x \rightarrow a} f(x) = L$$

$$f(a) ?$$

Methods of finding a limit:

- Direct substitution:

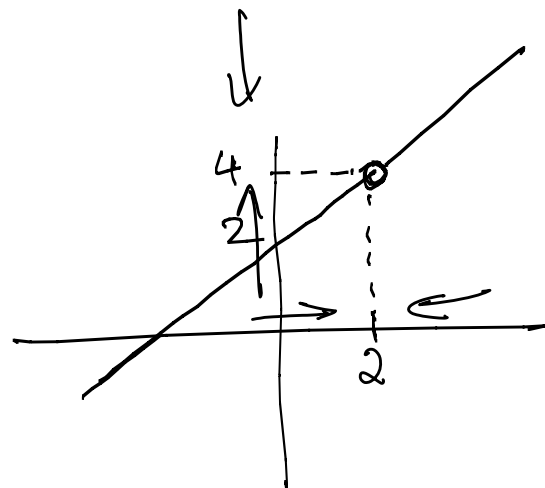
for "nice" functions.

$$\lim_{x \rightarrow 1} (x^2 + x - 2) = 0$$

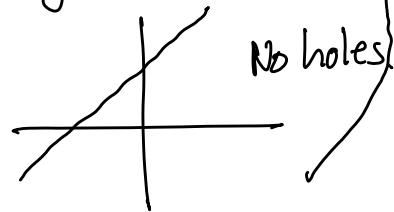
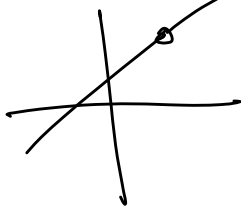
$$\lim_{x \rightarrow \pi/2} \sin x = 1$$

- Factorization

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \\ = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = 4 \end{aligned}$$



(Notice the difference $f(x) = \frac{x^2-4}{x-2}$ (see above) and $g(x) = x+2$)

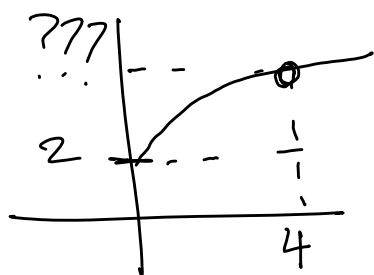


Eg. $f(x) = \frac{x-4}{\sqrt{x}-2}, x \neq 4$

Calculate $\lim_{x \rightarrow 4} f(x)$



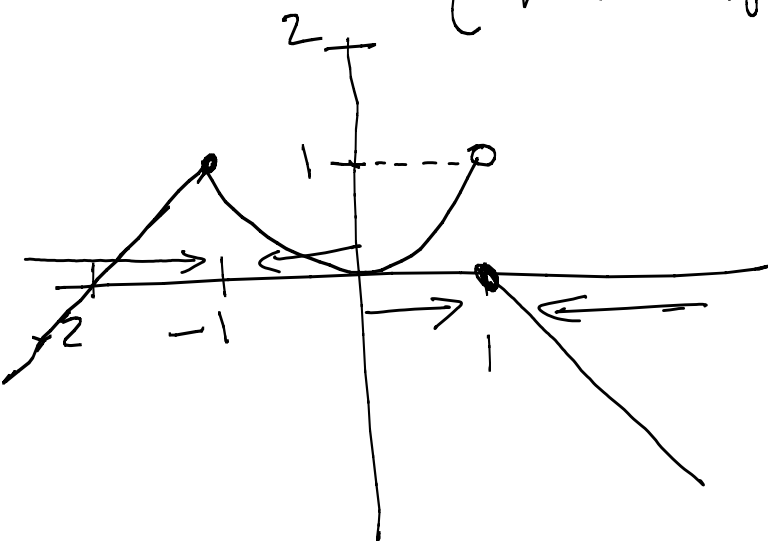
$$f(x) = \frac{x-4}{\sqrt{x}-2} = \frac{(\sqrt{x}-2)(\sqrt{x}+2)}{\sqrt{x}-2} = \sqrt{x}+2, x \neq 4$$



$$\lim_{x \rightarrow 4} f(x) = 4.$$

• One-sided limits

Eg $f(x) = \begin{cases} x+2 & \text{if } x \leq -1 \\ x^2 & \text{if } -1 < x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$



$$\lim_{x \rightarrow -1^-} f(x) = 1$$

$$\lim_{x \rightarrow -1^+} f(x) = 1.$$

$$\Rightarrow \lim_{x \rightarrow -1} f(x) = 1.$$

$$\lim_{x \rightarrow 1^-} f(x) = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = 0$$

$$\lim_{x \rightarrow 1^+} f(x) = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = 0$$

$$\lim_{x \rightarrow 1} f(x) \text{ does not exist}$$

$$\lim_{x \rightarrow 0} f(x) = 0$$

Theorem:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

More examples of one-sided limits

1. $\lim_{x \rightarrow 1^+} \sqrt{x-1}$



2. $\lim_{x \rightarrow 1^-} \sqrt{x-1}$
does not exist.

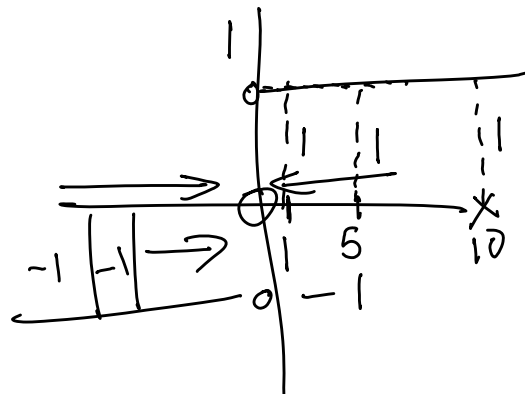
3. $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$

4. $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$

$\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x} & \text{if } x > 0 \\ -\frac{x}{x} & \text{if } x < 0 \end{cases}$$

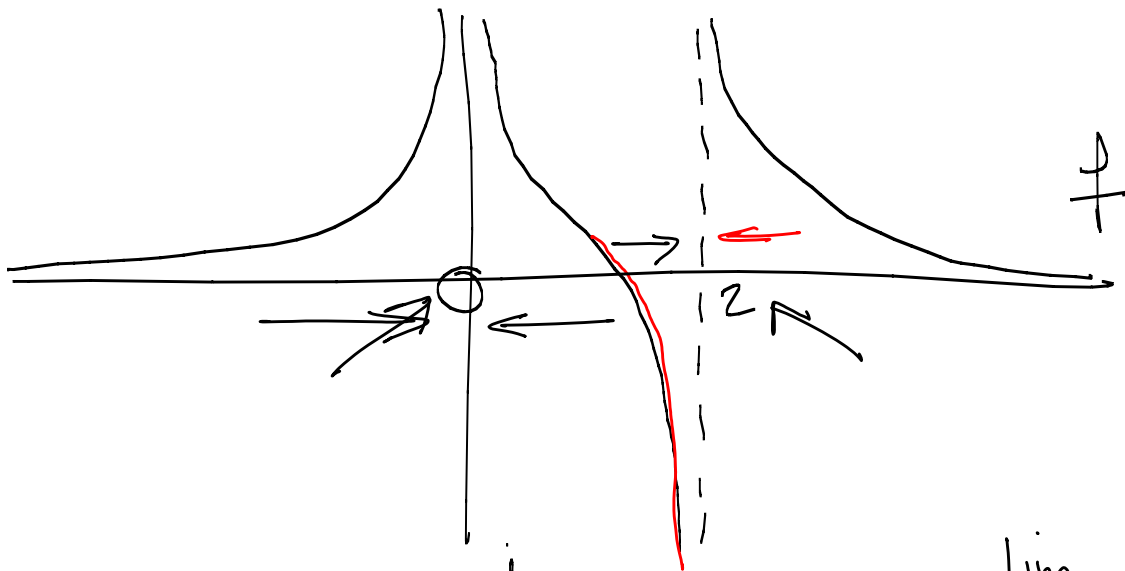
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Infinite limits

(We say such limits do not exist but write

$$\lim_{x \rightarrow a} f(x) = \infty \text{ for example})$$



$$\lim_{x \rightarrow 0^-} f(x) = \infty = \lim_{x \rightarrow 0^+} f(x) \Rightarrow \lim_{x \rightarrow 0} f(x) = \infty$$

$$\lim_{x \rightarrow 2^-} f(x) = -\infty \quad \lim_{x \rightarrow 2^+} f(x) = \infty$$

Format: $\frac{\text{number}}{\text{smaller and smaller}} \quad \left(\frac{c}{0}\right)$

Examples:

1. $\lim_{x \rightarrow 1^+} \frac{4}{x-1} = \infty$

~~Example 1~~

$\left(\frac{4}{0^+}\right)$

2. $\lim_{x \rightarrow 1^-} \frac{4}{x-1} = -\infty$

$\left(\frac{4}{0^-}\right)$

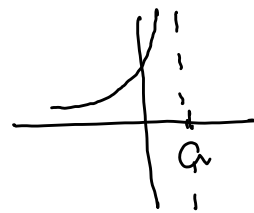
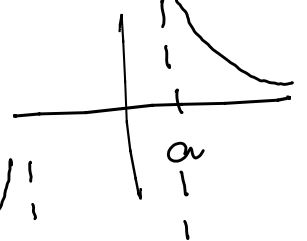
Vertical asymptotes

We say $x=a$ is a vertical asymptote (VA) of f if

$$\lim_{x \rightarrow a} f(x) = \infty \text{ (or } -\infty)$$

$$\lim_{x \rightarrow a^+} f(x) = \infty \text{ (or } -\infty)$$

$$\lim_{x \rightarrow a^-} f(x) = \infty \text{ (or } -\infty)$$



NB: $x=a$ cannot be in the domain of f .
so look for x -values not in the domain of f .

Examples:

Find the vertical asymptotes (if any) of:

1. $f(x) = x+1$ No VA.

2. $f(x) = \frac{x-1}{x-5}$

$$\lim_{x \rightarrow 5^+} \frac{x-1}{x-5} = \infty$$

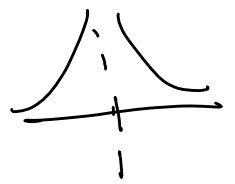
$$\lim_{x \rightarrow 5^-} \frac{x-1}{x-5} = -\infty$$

$$\left(\frac{4}{0^+}\right)$$

$$\left(\frac{4}{0^-}\right)$$

$\Rightarrow x=5$ is a VA.

$$3. f(x) = \frac{1}{(x-1)^2}$$



$$\lim_{x \rightarrow 1^+} \frac{1}{(x-1)^2} = \infty$$

$$\lim_{x \rightarrow 1^-} \frac{1}{(x-1)^2} = \infty$$

$\Rightarrow x=1$ is a VA.

$$4. f(x) = \frac{\sqrt{x}-3}{x-9} = \frac{\cancel{\sqrt{x}-3}}{(\cancel{\sqrt{x}-3})(\sqrt{x}+3)}, x \neq 9$$

$$\left(\frac{0}{0}\right)$$

$$\lim_{x \rightarrow 9} \frac{1}{\sqrt{x}+3} = \frac{1}{6}$$

